Appendix A. Supplementary Material

The severity shifts below are external inputs to the propagation step (Eq. (14)); the individual trajectories that follow correspond to the three high-responding patients discussed in the Results.

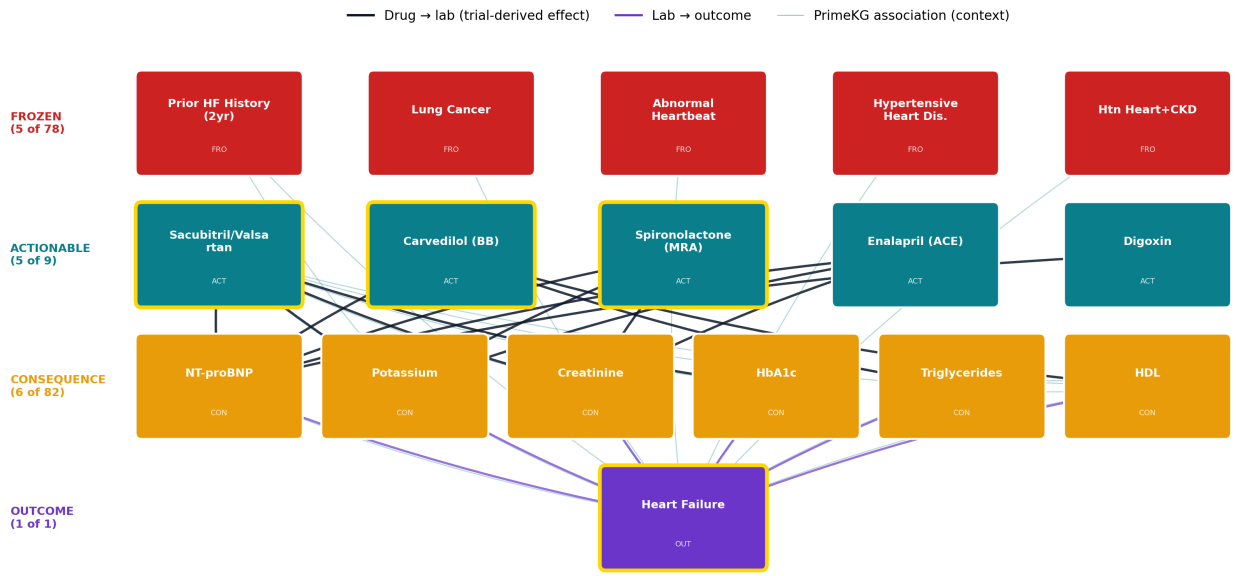


Fig. A1. Layered causal structure for the evaluated regimen, instantiating the four variable classes and the ordering of Eq. (1). Nodes are arranged by class from top to bottom: frozen pre-treatment characteristics (red), actionable drugs (teal), consequence laboratory mediators (orange), and the outcome (purple). Gold outlines mark the variables in the evaluated analysis: the angiotensin-receptor-neprilysin-inhibitor-based triple regimen (sacubitril/valsartan, carvedilol, and spironolactone), and incident heart failure. The identifying assumption is that treatment influences the outcome only through the laboratory mediators (treatment $\rightarrow$ labs $\rightarrow$ outcome). For legibility a representative subset of nodes is shown; the full graph derived from PrimeKG[15] contains 78 frozen, 9 actionable, and 75 consequence nodes. Additional actionable and consequence nodes that appear (e.g., antiplatelet and anticoagulant agents) belong to the broader knowledge graph and are not part of the evaluated regimens.

Table A1. Assumed twelve-month mediator shifts on the severity axis ($\Delta s_{d,\ell}$) for each candidate agent. Negative values denote a beneficial shift; positive values a worsening shift. Evidence grade: Q, a quantitative trial estimate is available for the mediator change; D, randomised trial evidence establishes the direction and relative ordering, with the magnitude calibrated onto the severity scale by the authors. Values are expressed on the normalised severity scale of Eq. (3); the mapping from reported clinical units is described in the Methods.

Agent	Mediator	$\Delta s_{d,\ell}$	Grade	Trial evidence	Source
Sacubitril/valsartan	NT-proBNP	−0.30	Q	28% median fall at 8–10 wk; 25% lower than enalapril at 4 wk and 8 mo	16,17
	HbA1c	−0.04	Q	Greater HbA1c fall than enalapril	18
	Potassium	+0.06	D	Hyperkalaemia 11.6% vs 14.0%	16
	Creatinine	+0.04	D	Renal dysfunction 10.1% vs 11.5%	16
Enalapril	NT-proBNP	−0.24	Q	Active comparator arm, PARADIGM-HF	16,17
	Potassium	+0.08	D	Hyperkalaemia 14.0%	16
	Creatinine	+0.06	D	Renal dysfunction 11.5%	16
Carvedilol	NT-proBNP	−0.20	D	Marked fall with carvedilol, not placebo	25
	Triglycerides	+0.06	D	Within-arm rise; lower than metoprolol	22
Spironolactone	Potassium	+0.12	Q	Median rise 0.3 mmol/L vs control	23,24
	Creatinine	+0.06	D	Serial creatinine monitoring mandated	23

Table A2. Change in mean predicted risk

Mediator	Severity shift	Δ mean risk (pp)
NT-proBNP	−0.66	−1.075
Potassium	+0.08	0.000
Creatinine	+0.10	0.000
HbA1c	−0.04	0.000
Triglycerides	+0.06	+0.014
HDL	+0.04	0.000

Table A3. Estimated twelve-month average treatment effect (ATE) on incident heart-failure risk by candidate regimen.

Regimen	Eligible (N)	ATE (pp)
ARNi-based triple (sacubitril/valsartan + BB + MRA)	188	−1.038
ACE-based triple (enalapril + BB + MRA)	202	−0.856
Sacubitril/valsartan	234	−0.502
Enalapril	309	−0.402
Carvedilol	280	−0.361
Spironolactone	307	−0.278

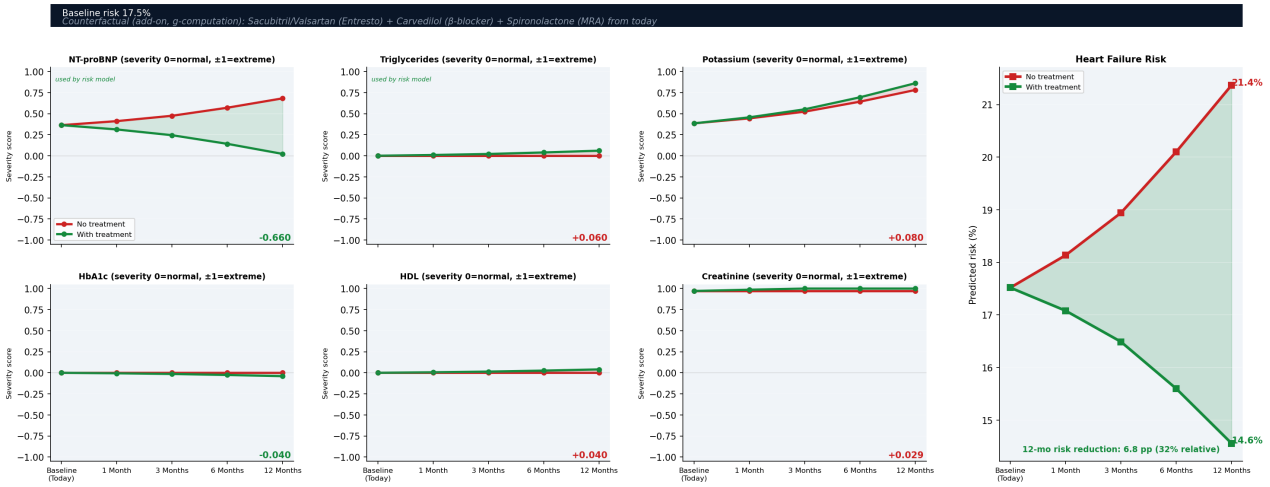


Fig. A2. Individual sequential counterfactual risk trajectories for a patient with a baseline predicted risk of 17.5%.



Fig. A3. Individual sequential counterfactual risk trajectories for a patient with a baseline predicted risk of 25.2%.



Fig. A4. Individual sequential counterfactual risk trajectories for a patient with a baseline predicted risk of 11.5%.